

Chapter-THREE:

Risk-Return and the Concept of Portfolio

Two Sides of the Investment Coin.

Chapter Three-Part-I

Basics of Risk and Return

(Single Asset Prospective)

Learning Objectives

- *After you read this chapter, you should be able to answer the following questions:*
 - ✓ What do we mean by *risk*, and what are some of the *alternative measures* of risk used in investments?
 - ✓ What do we mean by risk and *risk aversion*?
 - ✓ How do you compute the expected rate of return for an **individual risky asset**

Con't...

- ✓ How do you compute the standard deviation of rates of return for an individual risky asset?
- ✓ Understand the concept of portfolio
- ✓ Undertake Portfolio risk-return analysis of two or more securities
- ✓ Discuss the concepts of correlation and diversification

Con't...

- Risk analysis can be confusing, but it will help if you remember the following:
 1. All financial assets are expected to produce *cash flows*, and the risk of an asset is judged in terms of the risk of its cash flows.
 2. The risk of an asset can be considered in two ways: (1) on a *stand-alone basis*, or (2) in a *portfolio context*,
 3. In a portfolio context, an asset's risk can be divided into two components: (a) *diversifiable Risk*, and (b) *market risk*,

Con't...

- 4. An asset with a **high degree of relevant (market) risk** must provide a relatively **high expected rate of return** to attract investors.
 - Investors in general are *averse to risk*, so they will not buy risky assets unless those assets have **high expected returns**.
- 5. In this chapter, we focus on *financial assets* such as **stocks and bonds**, but the concepts discussed here also apply to *physical assets* such as **computers, trucks, or even whole plants**.

What is Uncertainty ?

- Whenever you make a **financing or investment** decision, there is some **uncertainty** about the outcome.
- *Uncertainty means not knowing exactly* what will happen in the future.
- There is uncertainty in most everything we do as financial managers, because no one knows precisely what changes will occur in such things as
 - tax laws,
 - consumer demand,
 - the economy, or
 - interest rates etc.

Con't....

- Though the terms “**risk**” and “**uncertainty**” are often used to mean the same thing, there is a distinction between them.
- **Uncertainty** is not knowing what’s going to happen. ***Risk is how we characterize how much uncertainty exists: The greater the uncertainty, the greater the risk.***
 - Risk is the degree of uncertainty.

Con't...

Risk Analysis

Risk --- What is this?

- **Consider the two cases.**
 - 1) Mr Ramesh has put his money in National Bank of Ethiopia (NBE) bond where **he is going to get 12% p.a.** He is really happy with the rate of return. Will he have sleepless nights, if the economy goes into **recession**? **Of course no.**
 - 2) Mr. Ramesh is very bullish with the stock market **and invests money into equity diversified fund** with the expectation **that he will get 12% return.** Will he have sleepless nights if economy goes into deep recession, and now he feels that he may get **negative returns of say 5-7%**? **Of course yes.**
- In the second situation, he has a fear, which is the result of huge difference in his expected return and the actual return, which he may get. This difference itself is the **risk** that he bears. Does he face this kind of difference in the first situation? **No. So there is no risk.**

What is risk ?

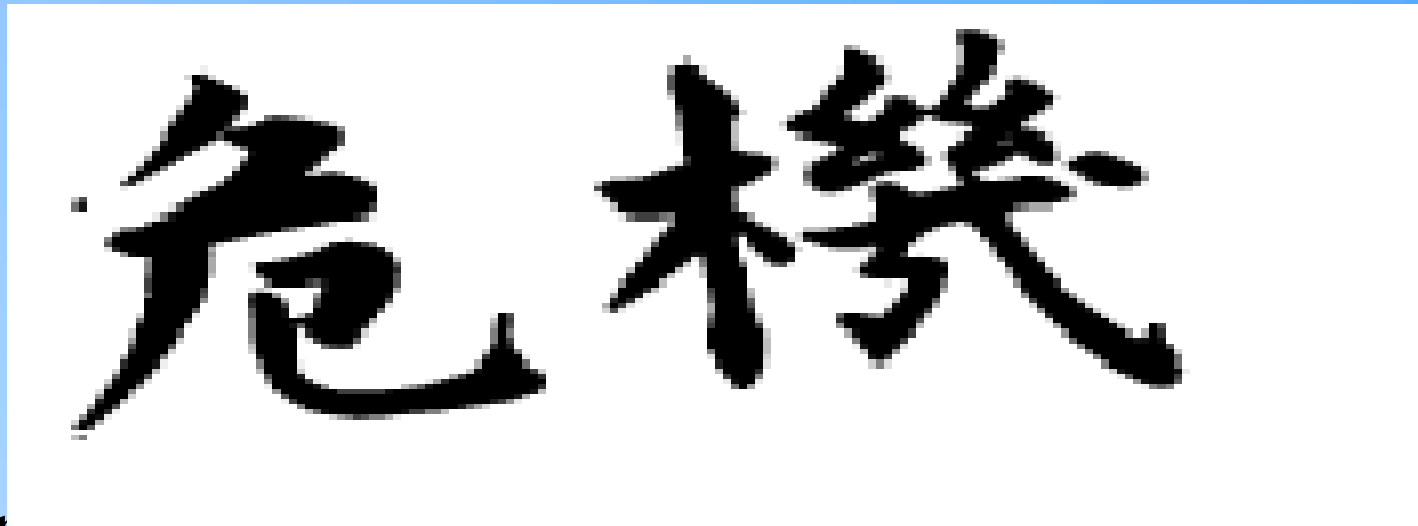
- Literally **risk** is defined as “exposing to **danger** or **hazard**” .
 - ✓ Which is perceived as **negative terms**.
- ✓ **In finance,**
 - **Risk** refers to the likelihood that we will receive a return on an investment that is **different** from the return we expected to make.
 - **Risk** is the probability or likelihood that **actual results (rates of return)** deviates from **expected returns**.

Con't...

- Thus, **risk** includes not only the **bad outcomes** (returns that are lower than expected), but also **good outcomes** (returns that are higher than expected).
- In fact, we can refer to the former as **downside risk** and the latter as **upside risk**.

What is Risk ?

- *Chinese Symbol for Risk:* The first symbol is the symbol for “**danger**” while the second is the symbol for “**opportunity**”, making risk a mix of danger and opportunity.

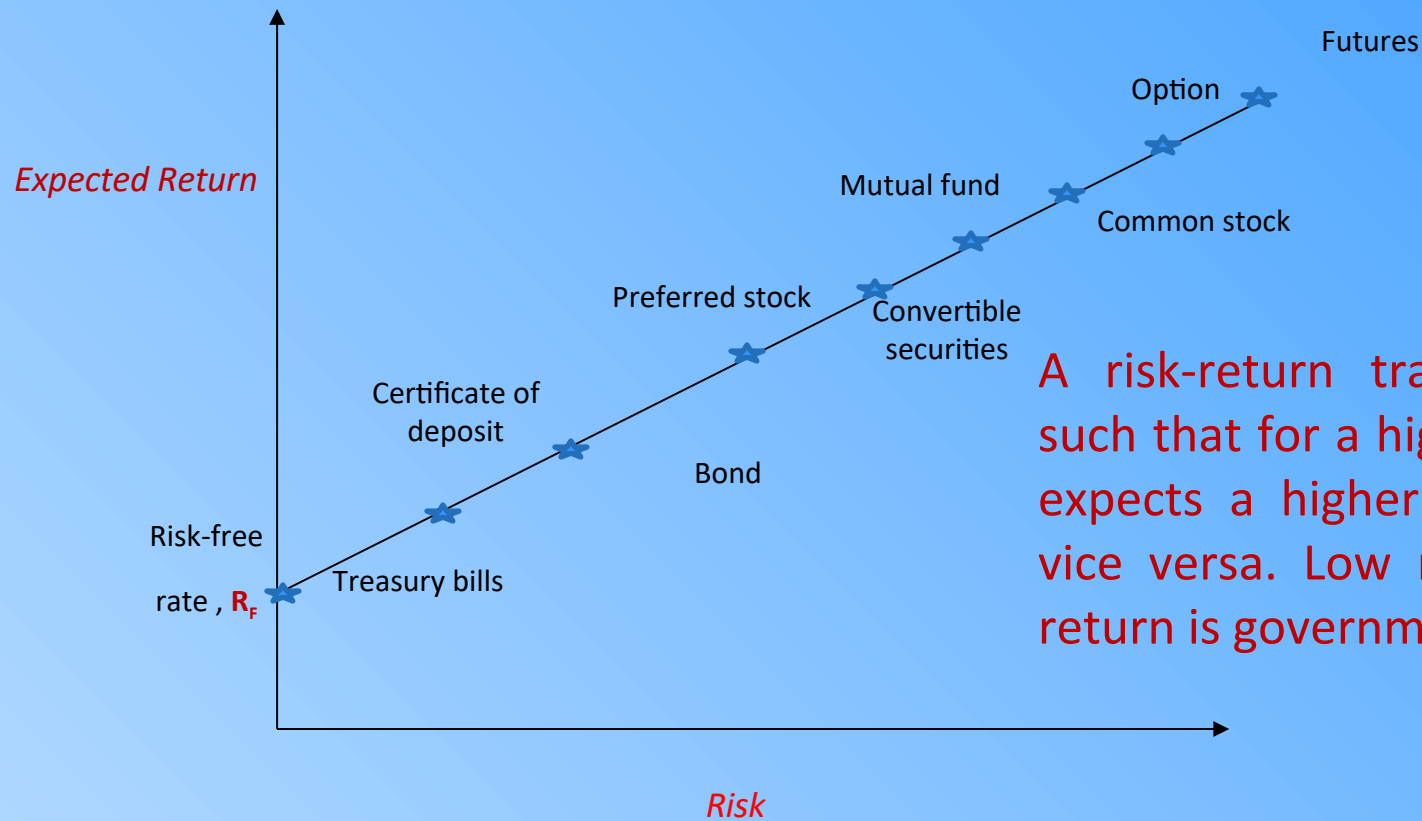


- Hence, **risk** is both bad outcomes and good outcomes.

Risk and Risk Premium

- **Risk** is the possibility that we won't achieve our expectations or the chance that actual investment returns will differ from those expected.
- Positive relationship – high risk; high return
- low risk; low return
- The relationship between risk and return is called **risk-return tradeoff**.

Risk-Return Tradeoffs for Various Investment Vehicles



A risk-return tradeoff exists such that for a higher risk one expects a higher return, and vice versa. Low risk and low return is government t-bill.

Sources of Risk

Sources of Risk Con't...

✓ **Business Risk:**

- Uncertainty of income flows caused by the nature of a firm's business
- Sales volatility and operating leverage determine the level of business risk.

Source of risk con't...

✓ Financial Risk

- Uncertainty caused by the use of debt financing. (Level of Financial Leverage)
- Borrowing requires fixed payments which must be paid ahead of payments to stockholders.
- The use of debt increases uncertainty of **stockholder income** and causes an increase in the stock's risk premium.

Source of Risk Con't...

✓ Liquidity Risk

- Uncertainty is introduced by the **secondary market** for an investment.
 - How long will it take to convert an investment into cash?
 - How certain is the price that will be received?

Source of Risk con't...

- **Exchange Rate Risk:**
- Uncertainty of return is introduced by acquiring securities denominated in a currency different from that of the investor.
- Changes in exchange rates affect the investors return when converting an investment back into the “home” currency.

Source of Risk con't...

- **Country Risk:**
- Political risk is the uncertainty of returns caused by the possibility of a major **change in the political or economic environment** in a country.
- Individuals who invest in countries that have unstable political-economic systems must include a country risk-premium when determining their required rate of return

Source of Risk Con't...

- ✓ **Interest rate risk** is the chance that changes in interest rates will adversely affect a security's value.
- ✓ **Purchasing Power Risk** refers to the chance that changing price levels (inflation or deflation) will adversely affect investment returns.

Diversifiable and Nondiversifiable Risk

- Although there are many reasons why **actual returns** may differ from **expected returns**, we can group the reasons into two categories:
 - A. **Market wide/Systematic Risk/Non-Diversifiable and**
 - B. **Firm-specific/ Unsystematic Risk/Unique risk.**

A. SYSTEMATIC RISK

- **Systematic risk** - The risk inherent to the entire market or entire market segments is known as **systematic risk**.
- The portion of the variability of return of a security that is caused by **external factors**, is called systematic risk.
- **Risk due to**
 - Economic and political instability,
 - Economic recession,
 - Inflation
 - Change in Government policy
 - Change in interest rate policy
 - Corporate tax rate
 - Foreign exchange control
 - Natural calamities etc.
- Thus the variation of return in shares, which is caused by these factors, is called systematic risk.

affect the price of all shares.

✓ **Systematic risk** cannot be reduced through diversification. This risk can be **mitigated through hedging** or by using the correct asset allocation strategy.

- **What is 'Asset Allocation:**

- Asset allocation is an investment strategy that aims to balance **risk and reward** by apportioning a portfolio's assets according to an individual's goals, risk tolerance and investment horizon.
- The three main asset classes - equities, fixed-income, and cash and equivalents - have different levels of risk and return, so each will behave differently over time.

B. NON - SYSTEMATIC RISK (UNSYSTEMATIC)

The return from a security sometimes varies because of certain factors affecting *only the company issuing such security*.

Risk due to

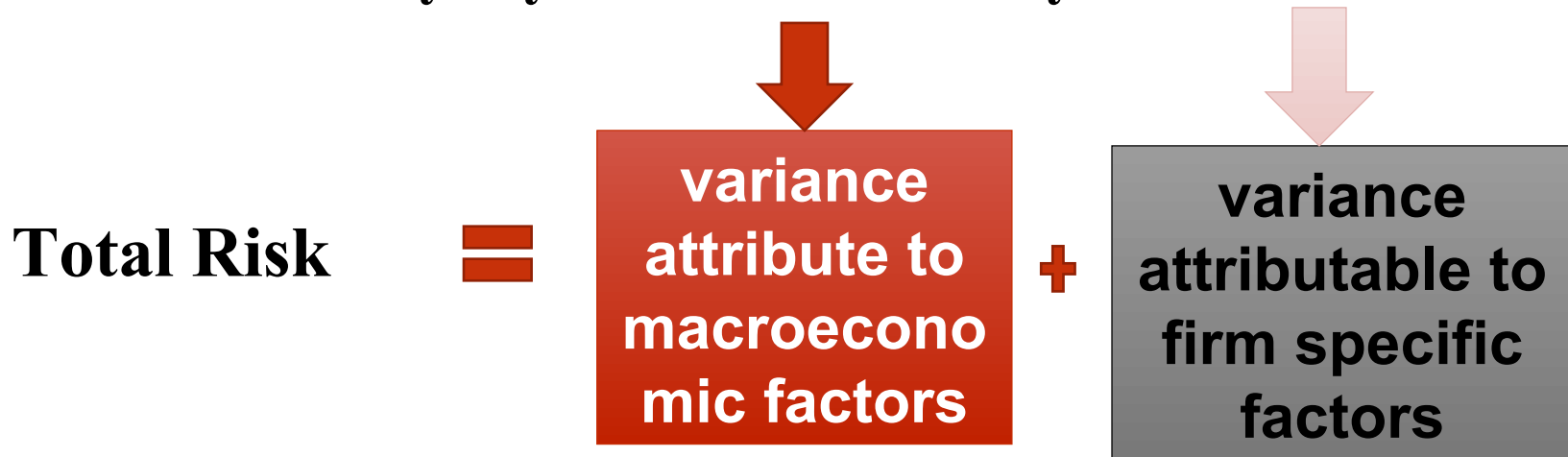
Shortage of raw material,
Labor strike,
R & D expert leave the company
Management inefficiency etc.

affect the share price of
one company.

When variability of returns occurs because of such firm-specific factors, it is known as unsystematic risk

Total risk

- Risk of an individual security is the variance (standard deviation) of its return.
- It consists of two parts:
 - **Total risk of a security = systematic risk + unsystematic risk**



Con't...

- Activity-1
- **Why Diversification Reduces or Eliminates Firm-Specific Risk: Give an Intuitive Explanation!**

Quantification of Returns and Risk

✓ Measuring of Return:

- If you buy an asset of any sort, your gain (loss) from that investment is called the **return on your investment**. This return will usually have **two components**:
- **Current return** – It is the periodic cash inflow in the form of **interest or dividend**.
- **Capital return** --- It represents change in the price of asset.
 - **Thus Total Return = Current Return + Capital Return**
- **Thus; return** is nothing but the reward for undertaking investment. Assessment of historical returns is must to know the performance of the fund manager. This also helps as an important input to estimate future returns.
- The **current return** can be **zero or positive**, whereas **capital return** can be **zero, positive or negative**.

Con't...

1. Calculation of Historical Returns (Ex post):

Single period return:

- **Example:** Suppose, at the beginning of the year, the stock for a company was selling for \$37 per share. If you had bought 100 shares, you would have a total out-lay of \$3700. Suppose, over the year, the stock paid you a dividend of \$1.85 per share. By the end of the year, then, you would have received income of:

– **Dividend Income in terms of dollar = $\$1.85 \times 100 = \185**

Con't...

- Also, suppose that the value of the stock has risen to \$40.33 per share by the end of the year. Your 100 shares are now worth \$4,033, so you have a **capital gain** of:

$$\text{Capital gain} = (\$40.33 - \$37) \times 100 = \$333$$

- Therefore, the total return of on your investment is the sum of the **dividend** and the **capital gain**.

$$\begin{aligned}\text{Total dollar return} &= \text{dividend income} + \text{capital gain (loss)} \\ &= \$185 + \$333 = \$ 518\end{aligned}$$

- **Notice** that, if you sold the stock at the end of the year, the total amount of cash you would have would equal your **initial investment** plus **total return**. Then, total cash if the stock is sold is :

Con't....

- Total cash = initial investment + total return
$$\$ 3700 + \$ 518 = \$ 4,218$$
- As a check, notice that this is the same as the **proceeds from the sale of the stock** plus **the dividends**:
- **Proceeds from stock sale + dividends = $\$40.33 \times 100 + 185 = \$ 4,218$**
- Although expressing returns in dollars is easy, two problems arise:
 - ✓ (1) to make a meaningful judgment about the return, you need to know the scale (size) of the investment;
 - ✓ A \$100 return on a \$100 investment is a **great return** (assuming the investment is held for 1 year), but a \$100 return on a \$10,000 investment would be a **poor return**.
 - ✓ **The question is, how much do we get for each dollar we invest?**

Con't...

- ✓ (2) You also need to know the **timing** of the return; a \$100 return on a \$100 investment is a great return if it occurs **after 1 year**, but the same dollar return after 20 years is **not very good**.
- The solution to these **scale** and **timing** problems is to express investment results as **rates of return**, or **percentage returns**.

Con't...

- **Basic Terms:**
- **Dividend yield:** The annual stock dividend as a percentage of the **initial stock price**.
- **Capital gains yield:** The **change in stock price** as a percentage of **the initial stock price**.
- **Total percent return:** The return on an investment measured as a percentage that accounts for all cash flows and capital gains or losses.

Con't....

- It is usually more convenient to summarize information about returns in **percentage terms**, rather than **in dollar terms**, because that way your return does not depend on how much you actually invest.
- To answer this question, let P_t be the price of the stock at the beginning of the year and let D_{t+1} be the dividend paid on the stock during the year.
- In the example above, the price at the beginning of the year was \$37 per share and the dividend paid during the year on each share was \$1.85. Therefore, dividend yield is:

Con't...

- **Dividend yield** = D_t / P_{t-1}
= $\$1.85 / 37 = .05 = 5\%$, this implies that for each dollar we invest, we get **five cents in dividends**.
- The second component of the return from investment is the capital gains yield. This is calculated as the change in the price during the year (the capital gain) divided by the beginning price:
- **Capital gains yield** = $(P_t - P_{t-1}) / P_{t-1}$
= $(40.33 - 37) / 37 = .09 = 9\%$. This means that per dollar we invest, we get nine cents in **capital gain**. Putting it together, per dollar invested, we get 5 cents in dividends and nine cents in capital gains: so we get a total of 14 cents. Our percentage return is 14%.
- Simply, the total percentage return of an investment can be calculated as:
- **The rate of return** = $((P_t + D_t - P_{t-1}) / P_{t-1}) \times 100 = ((40.33 + 1.85 - 37) / 37) \times 100 = 14\%$
- **Total Percentage return** =
$$\frac{\text{Dividend paid at end of period} + \text{Change in market value over period}}{\text{Beginning market value}}$$

= $\$1.85 + (40.33 - 37) / 37 = 5.18 / 37 = .14 = 14\%$

Measuring the Expected Return and Risk of a Single Asset

1) Calculation of Expected Return (EX ANTE)

✓ When we talk about **expectations**, we talk about **probability**.

✓ The future or expected return of a security is uncertain; however it is possible to describe the future returns statistically as a probability distribution.

▪ **The mean of this distribution is the expected return.**

✓ The **expected return** of the investment is the **probability weighted average** of all the possible returns.

Con't....

- ✓ If the **possible returns** are denoted by X_i and the related **probabilities** are $P(X_i)$, expected return may be represented as and can be calculated as:

$$E(R_i) = \sum X_i P(X_i).$$

- It is the sum of the products of possible returns with their respective probabilities. Consider the example below.

Con't...

- **Expected value of return** for asset, $E(R_i)$: Expected rate of return is the return expected to be realized from an investment.

$$E(R_i) = p_1 r_1 + p_2 r_2 + \dots + p_n r_n$$

where,

$p_1, p_2, \dots, p_n$ = is the probability of the i^{th} out come.

$r_1, r_2, \dots, r_n$ = is the i^{th} possible out come (return).

n = number of outcomes considered

or

$$E(R_i) = \sum (R_j \times Pr_j)$$

Where R_j = return for the j^{th} outcome

Pr_j = probability of occurrence of the j^{th} outcome

Con't...

- **Example:** Mr. X is considering the possible rates of return (dividend yield plus capital gain or loss) that he might earn next year on a \$10,000 investment in the stock of either **Alpha Company** or **Beta Company**. The rates of return probability distributions for the two companies are shown here under:

State of the economy	Probability of the state economy	Rate of return if the state economy occurs	
		Alpha Co	Beta Co.
Boom	0.35	20%	24%
Normal	0.40	15%	12%
Recession	0.25	5%	8%

Required: compute the expected rate of return on each company's stock and recommend where Mr "X" has to invest the \$10,000 investable fund.

Solution

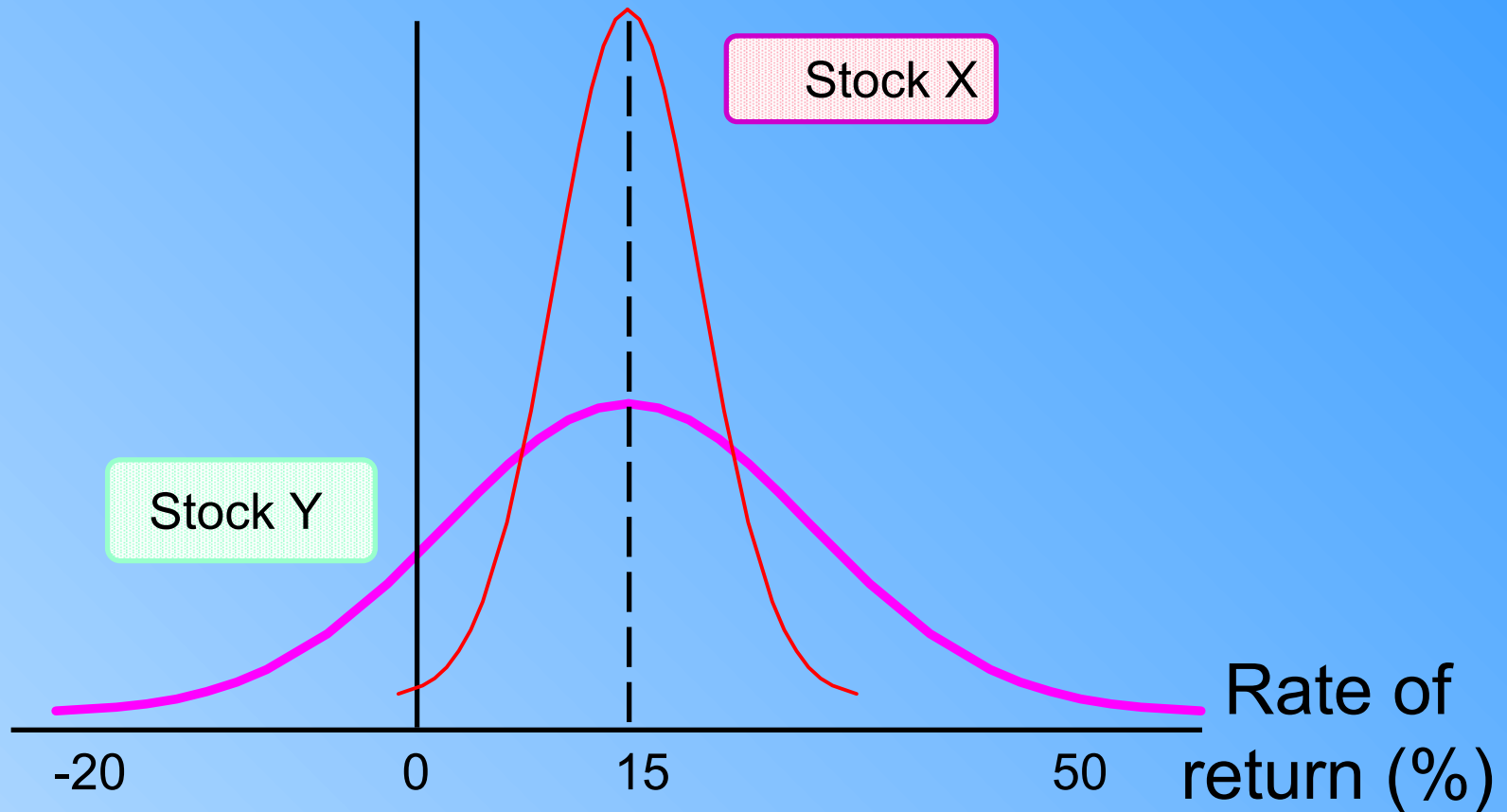
- $E(R_j) = \sum (R_j \times Pr_j)$
- $E(R_{\alpha}) = (0.35 * 20) + (0.4 * 15) + (0.25 * 5)$
 $E(R_{\alpha}) = 7 + 6 + 1.25 = \underline{14.25\%}$
- $E(R_{\beta}) = (0.35 * 24) + (0.4 * 12) + (0.25 * 8)$
 $E(R_{\beta}) = 8.4 + 4.8 + 2 = \underline{15.2\%}$

Calculation of Expected Risk

What is **investment** risk?

- Typically, investment returns are not known with certainty.
- **Investment risk** relates to the probability of earning a return less than **that expected**.
- The greater the chance of a return far below the expected return, the **greater the risk**.

Probability Distribution



■ Which stock is riskier? **Why?**

Answer: Con't...

- The tighter (or more peaked) the probability distribution, the more likely it is that the **actual outcome** will be close to the **expected value**, and hence the less likely it is that the actual return will end up far below the expected return.
 - Thus, the tighter the probability distribution, the **lower the risk assigned to a stock.**
- Since **Stock X** has a relatively tight probability distribution, its actual return is likely to be closer to its 15% expected return than that of **Stock Y**.
- According to this definition, Stock **X** is less risky than Stock **Y** because there is a smaller chance that its actual return will end up far below its expected return.

Measuring the Expected Risk of a Single Asset

- **Standard deviation** is the most common statistical indicator of an asset's risk (stand alone risk).
- **S.D** measures the variability of a set of observations.
- The larger the standard deviation, the higher the probability that actual returns will be **far below** the expected return.
- **Coefficient of variation** is an alternative measure of stand-alone risk.

Con't...

- **Standard deviation** is indicator of risk asset (an absolute measure of risk) of that asset's expected return, $\sigma (R_i)$, which measures the dispersion around its expected value.
- The **standard deviation** considers the distance (deviation) of each possible outcome from the expected value and the probability associated with that distance.
- This can be calculated using equation below:
$$\sigma (R_i) = \sqrt{\sum [R_j - E(R_i)]^2 \times Pr_j}$$

Con't...

- Steps to calculate the σ or sigma:

1. Calculate the expected rate of return:

Expected rate of return, $E(R_i) = \sum (R_j \times Pr_j)$

2. Subtract the expected rate of return ($E(R_i)$) from each possible outcome (r_i) to obtain a set of deviations about $E(R_i)$, $\text{Deviation}_i = r_i - E(R_i)$

Con't...

3. Square each deviation:

$$\text{Deviation}_i^2 = (r_i - E(R_i))^2$$

4. Multiply the squared deviations by the **probability of occurrence** for its related outcome.

$$P_i(r_i - E(R_i))^2$$

5. Sum these products to obtain the *variance* of the probability distribution:

6. $\text{Standard Deviation } (\sigma) = \sqrt{\text{Variance } (\sigma^2)}$

How to Determine the Expected Return and Standard Deviation

Stock BW			The expected return, <u>R</u>, for Stock BW is .09 or 9%
R_i	P_i	$(R_i)(P_i)$	
-.15	.10	-.015	
-.03	.20	-.006	
.09	.40	.036	
.21	.20	.042	
.33	.10	.033	
<i>Sum</i>	<i>1.00</i>	<i>.090</i>	

How to Determine the Expected Return and Standard Deviation

Stock BW			
R_i	P_i	$(R_i)(P_i)$	$(R_i - \bar{R})^2(P_i)$
-.15	.10	-.015	.00576
-.03	.20	-.006	.00288
.09	.40	.036	.00000
.21	.20	.042	.00288
.33	.10	.033	.00576
<i>Sum</i>	<i>1.00</i>	<i>.090</i>	<i>.01728</i>

Determining Standard Deviation (Risk Measure)

$$\sigma = \sqrt{\sum_{i=1}^n (R_i - \bar{R})^2 (P_i)}$$

Standard Deviation, σ , is a statistical measure of the variability of a distribution around its mean.

It is the square root of variance.

Note, this is for a discrete distribution.

Comments on Standard Deviation as a Measure of Risk

- Standard deviation (σ_i) measures total, or stand-alone, risk.
- The larger σ_i is, the lower the probability that actual returns will be **closer to expected returns**.
- Larger σ_i is associated with a wider **probability distribution of returns**.
- The larger standard deviation (σ_i) indicates a **greater variation of returns** and thus a *greater chance that the expected return will not be realized*.
- The larger the Standard deviation (σ_i), the higher the risk, because Standard deviation (σ_i), is a *measure of total risk*.

Coefficient of Variation: *A Relative Measure of Risk*

- If conditions for **two or more** investment alternatives are **not similar**—that is, if there are major differences in the *expected rates of return or standard deviation*—it is necessary to use a measure of *relative variability to indicate risk per unit of expected return*.
- The **coefficient of variation** is a useful measure of risk when we are comparing the investment alternatives which have
 - (i) same standard deviations but different expected values, or
 - (ii) different standard deviations but same expected values, or
 - (iii) different standard deviations and different expected values.

- *A widely used relative measure of risk is the coefficient of variation (CV), calculated as follows:*

- Formula for CV is:

$$\text{Coefficient of Variation} = \frac{\text{Standard Deviation}}{\text{Average or Expected Return}}$$
$$\text{CV} = \frac{\sigma(R_i)}{E(R_i)}$$

Coefficient of Variation *con't...*

CAse-1:

$$E(R_x)=10\%$$

$$S.D_x=8\%$$

$$E(R_y)=12\%$$

$$S.D_y=8\%$$

CAse-2:

$$E(R_x)=12\%$$

$$D_x=8\%$$

$$E(R_y)=12\%$$

$$S.D_y=10\%$$

CAse-3:

$$E(R_x)=7\%$$

$$D_x=5\%$$

$$E(R_y)=12\%$$

$$S.D_y=7\%$$

Chapter-THREE

Part-II

Risk_Return Analsys for Portfolio Assets

Introduction

- **Risk–return** trade-offs have an important role to play in corporate **finance theory** – from both a company and an investor perspective.
- Companies face variability in their **projected cash flows** whereas investors face variability in their **capital gains and dividends**.
- Assuming that companies and shareholders are rational, their aim will be to minimise the risk they face for a given return they expect to receive.
- So in this chapter we will examine how investors, by **‘not putting all their eggs in one basket’**, are able to reduce the risk they face given the level of their expected return.

Con't...

- **PORTFOLIO**: is a collection or a group of investment assets. If you hold only one asset, you suffer a loss if the return turns out to be very low.
- If you hold two assets, the chance of suffering a loss is reduced and returns on both assets must be low for you to suffer a loss.
- By diversifying, or investing in multiple assets that do not move proportionately in the same direction at the same time, you **reduce your risk**.

Con't...

- **Portfolio analysis** is used to determine the **return and risk** for these combinations of assets.
- It is the **total portfolio risk** and **return** that is important.
- The risk and return of individual assets should not be analyzed in isolation; rather they should be analyzed in terms of how they affect the **risk and return of the portfolio** in which they are included.
- The goal of the financial manager should be to create an efficient portfolio, one that **maximizes return for a given level of risk or minimizes risk for a given level of return.**

Portfolio Theory

Two Approaches to Constructing Portfolios

I. Traditional Approach

versus

II. Modern Portfolio Theory

Traditional Portfolio Theory

- An investment advisor would have given you advice like:
 - *If you are young you should be putting money into a couple of good **growth stocks**, maybe even into a few small stocks.*
 - *If you're close to retirement, you should be putting all of your money into **bonds and safe stocks**, and nothing into the risky stocks*
- Tends to focus on well-known companies
 - Perceived as less risky
 - Stocks are more liquid and available
 - Familiarity provides higher “comfort” levels for investors

Modern Portfolio Theory (MPT)

- It deals with the selection of **optimal portfolios** by rational risk averse investors-i,e by investors who attempt to maximize their expected portfolio return consistent with individually acceptable levels of portfolio risk.
 - Given a choice between two assets with equal rates of return, risk-averse investors will select the asset with the lower level of risk.
- **A risk-averse investor will prefer**
 - a portfolio with the highest expected return for a given level of risk or
 - prefer a portfolio with the lowest level of risk for a given level of expected return. In portfolio theory, this is referred to as the *principle of dominance*.
- Portfolio theory determines the combination of risk and return that allows the investors to achieve the highest return for a given level of risk.

Modern Portfolio Theory (MPT)

- Emphasizes statistical measures to develop a portfolio plan
- **Focus is on:**
 - Expected returns
 - Standard deviation of returns
 - Correlation between returns

Markowitz Portfolio Theory

■ Quantify risk

- ✂ In the early 1960s, the investment community talked about **risk of the portfolio**, but there was no specific measure for the risk of the portfolio.
- ✂ To build a portfolio model, however, investors had to quantify their **risk variable**.
- ✂ **Portfolio theory** , originally proposed by **henry markowitz** in the 1952, was the first formal attempt to quantify the risk of a portfolio and develop a methodology for determining the optimal portfolio.
 - He was the first person to show **quantitatively why and how diversification reduces risk**.
 - Derives the expected rate of return for a portfolio of assets and an expected risk measure
 - Derives the formula for computing the variance of a portfolio, showing how to effectively diversify a portfolio

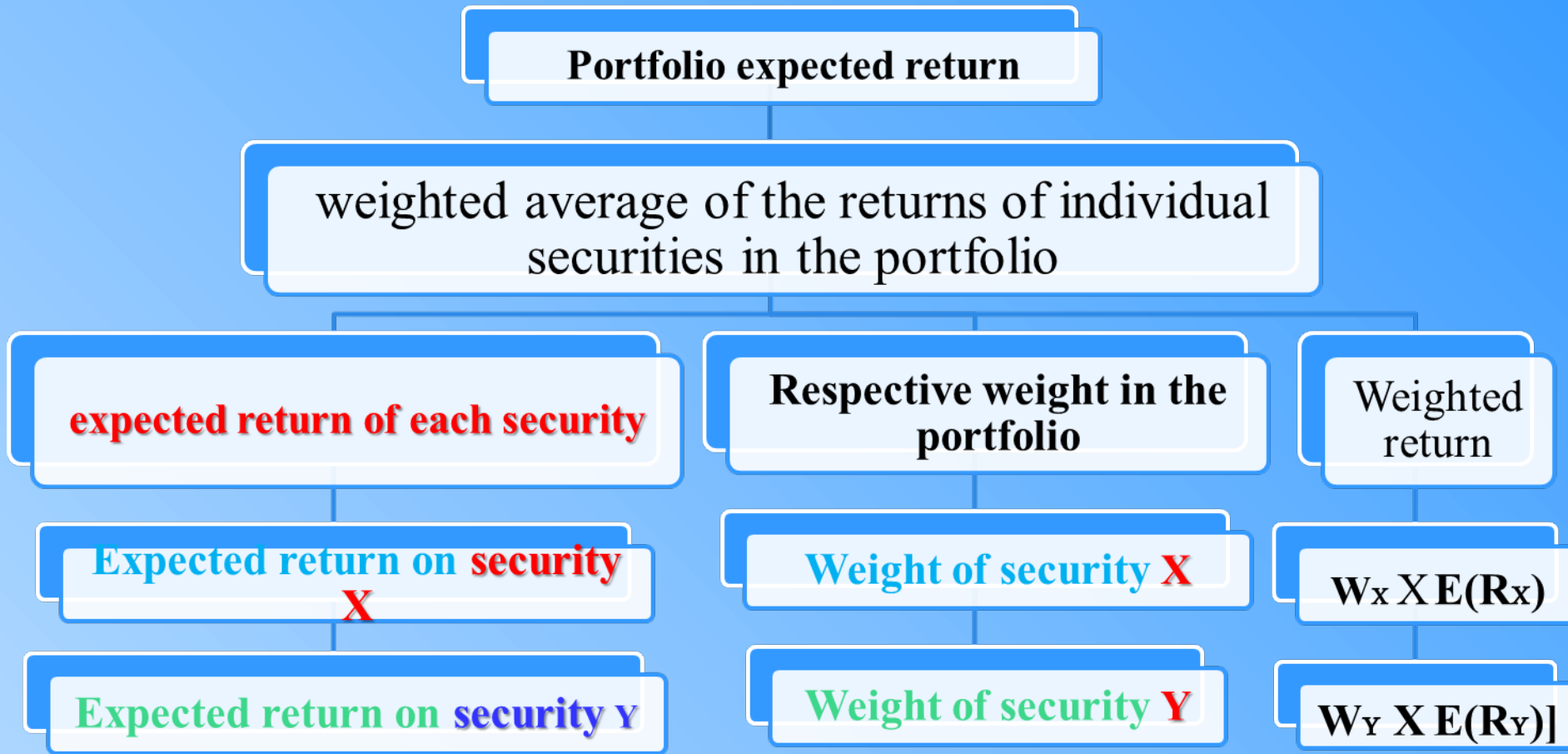
PORTFOLIO RETURN: TWO-ASSET CASE

↗ The return of a portfolio is equal to the **weighted average of the returns of individual assets** (or securities) in the portfolio.

↗ Requirement

- Expected return of individual security
- Portfolio weight
 - Portfolio weights: proportion of fund invested in each security / Percentages invested in each asset (w_i) serve as the weights
- Multiply by respective weight and sum the result

Expected return of a portfolio



$$E[r] = P_1 r_1 + P_2 r_2 + P_3 r_3 + \dots P_n r_n$$

$$= \sum_{i=1}^n P_i r_i$$

$$E(R_p) = [W_X \times E(R_X)] + [W_Y \times E(R_Y)] + \dots [W_n \times E(R_n)]$$

$$E(R_p) = \sum_{i=1}^n W_i E(R_i)$$

Con't....

- The **expected return** on a portfolio, $\hat{r}_p$, is simply the weighted average of the expected returns on the individual assets in the portfolio, with the weights being the fraction of the total portfolio invested in each asset:

$$\hat{r}_p = w_1\hat{r}_1 + w_2\hat{r}_2 + \dots + w_n\hat{r}_n$$

Formula : $E(R_p) = \sum_{i=1}^n W_i R_i$

Where:

- ✓ R_p = Expected Return of the Portfolio.
- ✓ W_i = The weights attached for each security.
- ✓ $\hat{r}_1, \hat{r}_2, \hat{r}_3, \dots, \hat{r}_n$ = Expected return of a security.
- ✓ n = No of stocks/ securities in a portfolio.

Con't...

- Note; “ W_i ” is the fraction of the portfolio’s dollar value invested in stock/security i , that is, the value of investment in stock i divided by the total value of the portfolio. Thus, the sum of W_i ’s must be 1.00.
- **Example:** Consider a portfolio of three stocks A, B, and C, with expected returns of 16%, 12%, and 20% respectively. The portfolio consists of 50% stock A, 25% stock B, and 25% stock C.
- **Required:** what is the expected return on this portfolio?
- $E(R_p) = .16*50 + .12*25 + .20*25 = 16\%$ which is the expected return of a portfolio

Con't...

Suppose we have the following projections on three stocks:

<i>State of Economy</i>	<i>Probabilities</i>	<i>Returns if state occurs</i>		
		<i>Stock A</i>	<i>Stock B</i>	<i>Stock C</i>
<i>Boom</i>	<i>0.40</i>	<i>10%</i>	<i>15%</i>	<i>20%</i>
<i>Bust</i>	<i>0.6</i>	<i>8%</i>	<i>4%</i>	<i>0%</i>

We want to calculate portfolio expected returns in two cases. First, what would be the expected return on a portfolio with equal amounts invested in each of the three stocks? Second, what would be the expected return if half of the portfolio were in A, with the remainder equally divided between B and C? (8pt)

Another Example

- Consider the following information

– State	Probability	X	Z
– Boom	.25	15%	10%
– Normal	.60	10%	9%
– Recession	.15	5%	10%

- What is the expected return and standard deviation for a portfolio with an investment of \$6000 in asset X and \$4000 in asset Y?

PORTFOLIO EXPECTED RETURN Con't...

- **Example-2:** A portfolio consists of four securities with expected returns of 12%, 15%, 18%, and 20% respectively. The proportions of portfolio value invested in these securities are 0.2, 0.3, 0.3, and 0.20 respectively.

$$E(R_p) = \sum_{i=1}^n w_i E(R_i)$$

where $E(R_p)$ = expected portfolio return

w_i = weight assigned to security i

$E(R_i)$ = expected return on security i

n = number of securities in the portfolio

Then expected return on the portfolio of the above Example is:

$$\begin{aligned} E(R_p) &= 0.2(12\%) + 0.3(15\%) + 0.3(18\%) + 0.2(20\%) \\ &= \mathbf{16.3\%} \end{aligned}$$

Ex Ante Portfolio Returns

Simply the Weighted Average of Expected Returns

Example-3:

	Relative Weight	Expected Return	Weighted Return
Stock X	0.400	8.0%	0.03
Stock Y	0.350	15.0%	0.05
Stock Z	0.250	25.0%	0.06
Expected Portfolio Return =			<u><u>14.70%</u></u>

Risk in a Portfolio Context

- **As** we just saw, the expected return on a portfolio is simply the weighted average of the expected returns on the individual assets in the portfolio.
- However, unlike returns, the risk of a portfolio, **standard deviation of a portfolio**, is generally *not* the weighted average of the standard deviations of the individual assets in the portfolio; the portfolio's risk will almost always be *smaller* than the weighted average of the assets' S.D's.
- Markowitz showed how the risk of a portfolio is **NOT** just the weighted average sum of the risks of the individual securities...but rather, also a function of the degree of co movement of the returns of those individual assets.

- **Portfolio risk** can be reduced by the simple kind of diversification, that is; when different assets are added to the portfolio, the **total risk tends to decrease**.
- **Total risk of portfolio** consists of **systematic risk & unsystematic risk** and this total risk is measured by the **variance** and **standard deviation** of the rate of return overtime.
- As the **portfolio size increases**, the **total risk decreases up to a certain** level. Beyond that limit, risk cannot be reduced.
 - This indicates that spreading out the assets beyond certain level can't be expected to reduce the portfolio's total risk below the level of **un-diversifiable risk**.
- Markowitz showed that the **variance of the rate of return** was a meaningful measure of portfolio risk under a reasonable set of **assumptions**, and he derived the formula for computing the **variance of a portfolio**.

Assumptions of the Markowitz Model:

- **The Markowitz model is based on several assumptions regarding investor behaviour:**
 1. Investors consider each investment alternative as being represented by a **probability distribution of expected returns** over some holding period.
 2. Investors maximize one-period expected utility, and their utility curves demonstrate diminishing marginal utility of wealth.
 3. Investors estimate the **risk of the portfolio on the basis of the variability of expected returns**.

Con't...

4. Investors base decisions solely on **expected return and risk**, so their utility curves are a function of expected return and the expected variance (or standard deviation) of returns only.
5. For a given **risk level**, investors prefer **higher returns to lower returns**. Similarly, for a given level of expected return, investors prefer less risk to more risk.

Con't...

- **Therefore:**

- ✓ Under these assumptions, a single asset or portfolio of assets is considered to be efficient if **no** other asset or portfolio of assets offers **higher expected return** with the same (or lower) risk, or lower risk with the **same (or higher) expected return**.

Diversifying unsystematic risk using a two-share portfolio

- The simplest portfolio to consider is that **containing two shares**.
- ***Diversification*** is the combination of assets whose returns do not vary with one another in the **same direction at the same time**.
- The extent to which a two-share portfolio will reduce unsystematic risk depends on the **correlation** between the two shares' returns.
- This correlation is quantified by the **correlation coefficient (ρ)** of the returns of the two shares, which can take any value in the range **-1 to +1**.

Correlation

- **Correlation (co-movement)**: refers to the association of movement between **two numbers**.
- It measures the **degree** of linear relationship to which two variables, such as **returns on two assets, move together**.
- Correlation takes on numerical values that range from **+1 to -1**.

Con't...

- While the **positive or negative sign** indicates the **direction of the co-movement**.
- The closer the correlation coefficient is to +1 or -1, the stronger the association.
 - +1 is perfect positive correlation.
 - -1 is perfect negative correlation.
 - 0.0 no relationship
- If the variables move together, they are positively correlated (a positive correlation coefficient).
- If the variables move in opposite direction, they are negatively correlated (a negative correlation).

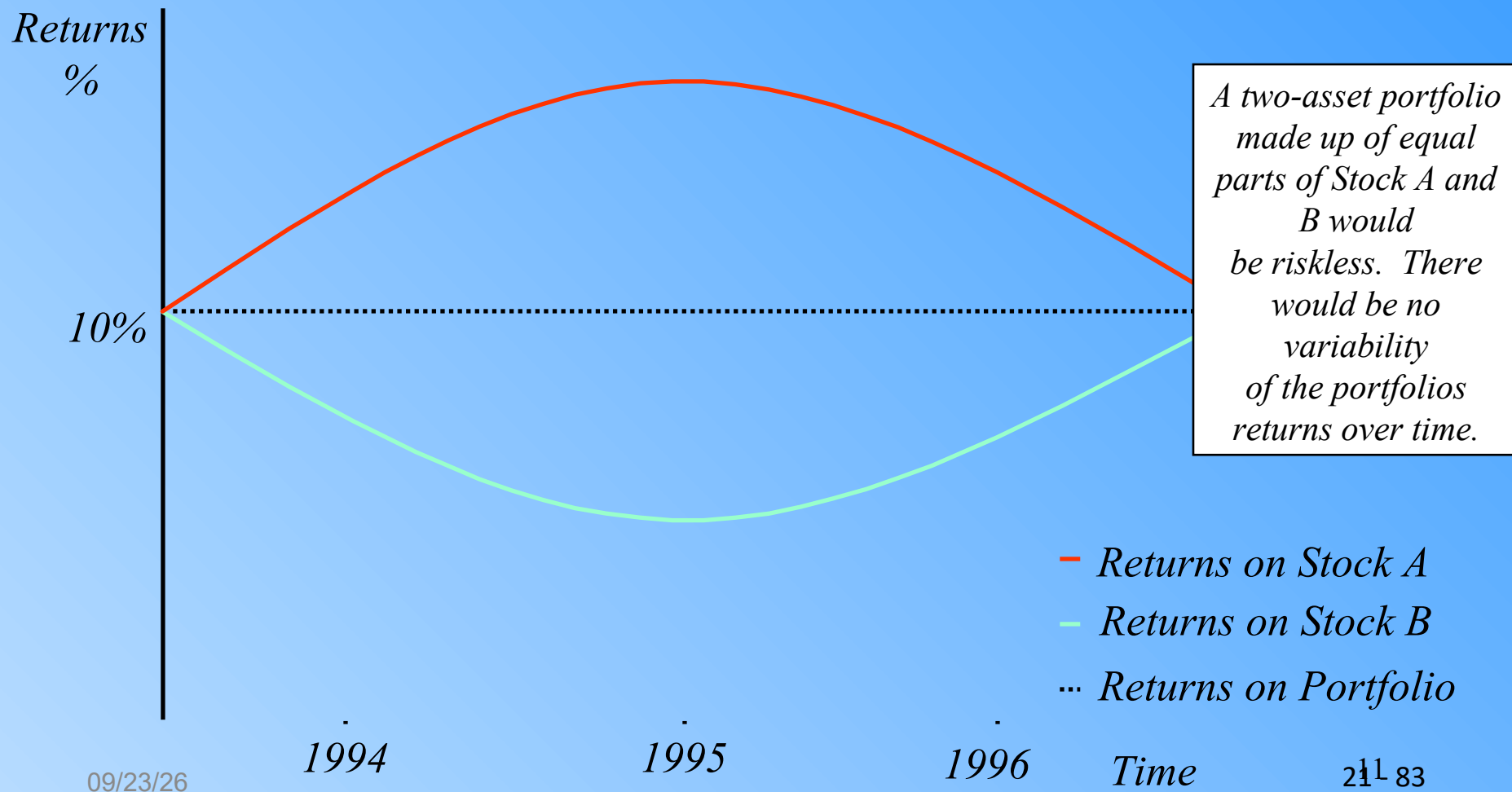
Con't...

- A correlation of **+1.0** indicates that the variables **move up** and **down together**, the relative magnitude of the movements is exactly the same (perfect positive correlation).
- A correlation of **-1.0** implies that they move exactly opposite to each other. (Perfect negative correlation).
- A correlation of **0.0** indicates that **no relationship** between the variables, that is they are unrelated.

Con't...

- If correlation is between 0.0 and +1.0, the returns usually **move up and down together**, but not all the time. The closer the correlation is to 0.0, the lesser the two sets of returns **move together**.
- A **correlation coefficient falling between -1 and 0** indicates negative, but not perfect negative correlation between the two assets' returns.

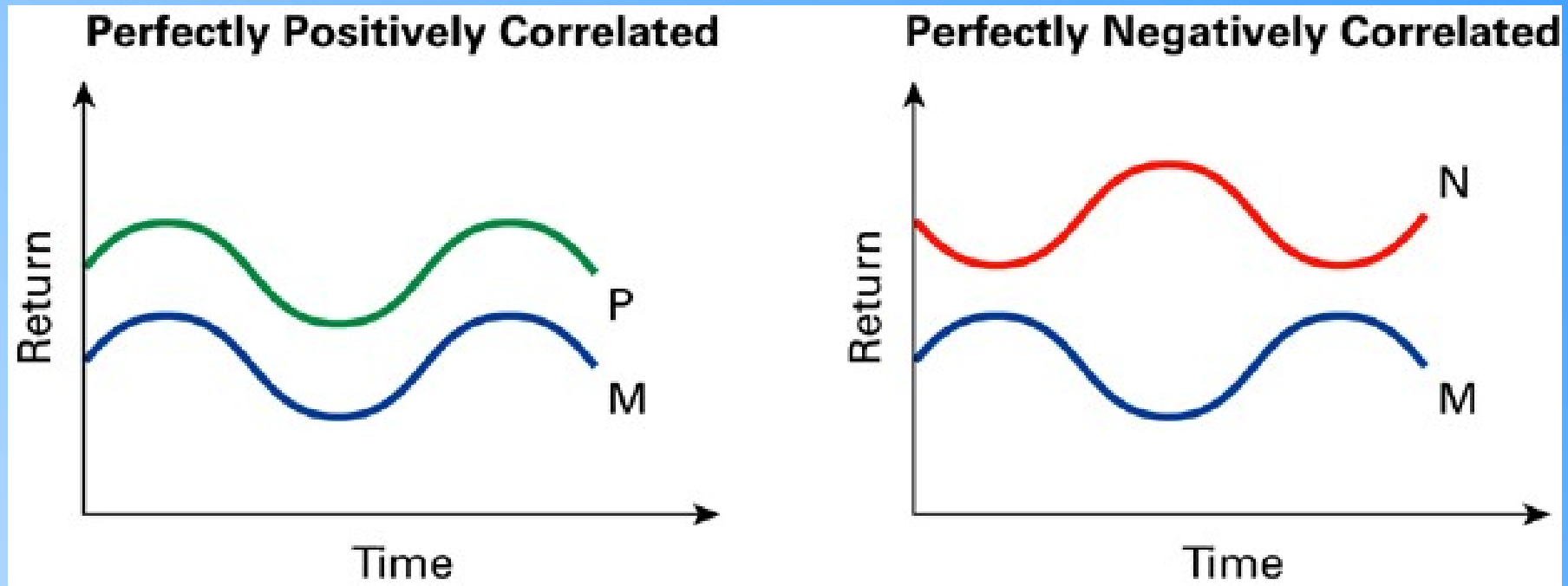
Perfect Negatively Correlated Returns over Time



Con't...

- ✓ If $\rho(A,B) = +1$ no unsystematic risk can be diversified away
 - ✓ If $\rho(A,B) = -1$ all unsystematic risk will be diversified away
 - ✓ If $\rho(A,B) = 0$ no correlation between the two securities' returns
-
- Therefore, when picking a two-share portfolio it is most beneficial to choose two shares whose **correlation coefficient** is as close to **-1** as possible.
 - However, as long as the correlation coefficient is **less than +1**, some unsystematic risk will be **diversified away**.

The Correlation Between Series M, N, and P



Correlation:

Why Diversification Works!

- To reduce overall risk in a portfolio, it is best to combine assets that have a *negative (or low-positive) correlation*.
- *Uncorrelated* assets reduce risk somewhat, but not as effectively as *combining negatively correlated assets*.
- When correlation coefficient of returns on individual securities is perfectly positive (i.e., $\text{cor} = 1.0$), then there is no advantage of diversification.
 - We may therefore conclude that diversification always reduces risk provided the correlation coefficient is less than 1.

Con't...

- In practice it is difficult to find two securities whose correlation coefficient is exactly **-1**, but the most commonly quoted example is that of an **umbrella manufacturer** and **an ice cream company**.
- The classical **example** of the benefit of diversification is to consider the effect of combining investment in **an ice-cream producer** with the investment in a **manufacturer of umbrella**.

Con't...

- For simplicity assume that the return to the ice-cream producer is **+15% if the weather is sunny** and **-10% if it is rain**. Similarly the manufacturer of umbrella benefits when it **rains (+15)** and **looses when the sun shines (-10%)**. Further assume that each of the two weather states occur with probability of 50%.
- *Thus, **diversification** does nothing to reduce risk if the portfolio consists of perfectly positively correlated stocks.*

Co-efficient of Correlation

$$\begin{aligned}\text{Cor } (R_i, R_j) \text{ or } \rho_{ij} &= \frac{\text{Cov } (R_i, R_j)}{\sigma_i \sigma_j} \\ &= \frac{\sigma_{ij}}{\sigma_i \sigma_j} \\ \sigma_{ij} &= \rho_{ij} \cdot \sigma_i \cdot \sigma_j\end{aligned}$$

where ρ_{ij} = correlation coefficient between the returns on securities i and j

σ_{ij} = covariance between the returns on securities
 i and j

σ_i, σ_j = standard deviation of the returns on securities

CO-VARIANCE OF RETURNS

- **Covariance** measures how closely security returns move together. The covariance between possible returns for securities J and K,
- When we consider two assets in the portfolio, we are concerned with the **co-movement** of security movement.
- **Co-movements** between the returns of securities are measured by **covariance (an absolute measure)** and **coefficient of correlation (a relative measure)**.
- It can be:
 - Positive Covariance,
 - Negative Covariance, and
 - Zero Covariance,

Con't...

- A **positive covariance** between the returns of two securities indicates that the returns of the two securities tend to move in the **same direction**.
- A **negative covariance** between the returns of two securities indicates that the returns of the two securities tend to move in **opposite directions**.
- A relatively **small or zero covariance** between the returns of two securities indicates that there is little or no relationship between the returns of the two securities;
- We denote the **covariance** between the return of security **i** and the return of security **j** by σ_{ij} (the Greek letter **sigma**, **σ_{ij}** or covariance **ij**).

COVARIANCE

- **Covariance will flow with the following steps:**
- **Step 1:** Calculate the expected return for each assets
- **Step 2:** For each scenario and investment, subtract the investment's expected return from its possible outcome.
- **Step 3:** For each scenario, multiply the deviations for the two investments.
- **Step 4:** Weight this product by the scenario's probability.
- **Step 5:** Sum these weighted products to arrive at **the covariance**.

Con't....

$$\begin{aligned} \text{Cov} (R_i, R_j) = & p_1 [R_{i1} - E(R_i)] [R_{j1} - E(R_j)] \\ & \vdots \\ & + p_2 [R_{i2} - E(R_i)] [R_{j2} - E(R_j)] \\ & + \\ & + p_n [R_{in} - E(R_i)] [R_{jn} - E(R_j)] \end{aligned}$$

ILLUSTRATION

The returns on assets 1 and 2 under five possible states of nature are given below

State of nature	Probability	Return on asset 1	Return on asset 2
1	0.10	-10%	5%
2	0.30	15	12
3	0.30	18	19
4	0.20	22	15
5	0.10	27	12

The expected return on asset 1 is :

$$E(R_1) = 0.10 (-10\%) + 0.30 (15\%) + 0.30 (18\%) + 0.20 (22\%) + 0.10 (27\%) = 16\%$$

The expected return on asset 2 is :

$$E(R_2) = 0.10 (5\%) + 0.30 (12\%) + 0.30 (19\%) + 0.20 (15\%) + 0.10 (12\%) = 14\%$$

The covariance between the returns on assets 1 and 2 is calculated below :

<i>State of nature</i>	<i>Probability</i>	<i>Return on asset 1</i>	<i>Deviation of the return on asset 1 from its mean</i>	<i>Return on asset 2</i>	<i>Deviation of the return on asset 2 from its mean</i>	<i>Product of the deviations times probability</i>
(1)	(2)	(3)	(4)	(5)	(6)	(2) x (4) x (6)
1	0.10	-10%	-26%	5%	-9%	23.4
2	0.30	15%	-1%	12%	-2%	0.6
3	0.30	18%	2%	19%	5%	3.0
4	0.20	22%	6%	15%	1%	1.2
5	0.10	27%	11%	12%	-2%	-2.2
						Sum = 26.0

Thus the covariance between the returns on the two assets is 26.0.

Grouping Individual Assets into Portfolios

- The riskiness of a portfolio that is made of different risky assets is a function of three different factors:
 - the riskiness of the individual assets that make up the portfolio
 - the relative weights of the assets in the portfolio
 - the degree of **co-movement** of returns of the assets making up the portfolio.
- The **standard deviation** of a **two-asset portfolio** may be measured using the Markowitz model:

$$\sigma_p = \sqrt{\sigma_1^2 w_1^2 + \sigma_2^2 w_2^2 + 2w_1 w_2 \rho_{1,2} \sigma_1 \sigma_2}$$

Expected Return and Risk For Portfolios

Standard Deviation of a Two-Asset Portfolio using Covariance

[8-11]

$$\sigma_p = \sqrt{\underbrace{(w_A)^2(\sigma_A)^2}_{\text{Risk of Asset A adjusted for weight in the portfolio}} + \underbrace{(w_B)^2(\sigma_B)^2}_{\text{Risk of Asset B adjusted for weight in the portfolio}} + \underbrace{2(w_A)(w_B)(COV_{A,B})}_{\text{Factor to take into account comovement of returns. This factor can be negative.}}}$$

Risk of Asset A
adjusted for weight
in the portfolio

Risk of Asset B
adjusted for weight
in the portfolio

Factor to take into
account comovement of
returns. This factor
can be negative.

Example

- + Determine the expected return and standard deviation of the following portfolio consisting of two stocks that have a correlation coefficient of .75.

Portfolio	Weight	Expected Return	Standard Deviation
Apple	.50	.14	.20
Coca-Cola	.50	.14	.20

Answer

□ Expected Return of the Portfolio = $.5 (.14) + .5 (.14) = .14$ or **14%**

□ Standard deviation of the portfolio

$$= \sqrt{(.5^2 \times .2^2) + (.5^2 \times .2^2) + (2 \times .5 \times .5 \times .75 \times .2 \times .2)}$$
$$= \sqrt{.035} = .187 \text{ or } \mathbf{18.7\%}$$

➤ Lower than the weighted average of 20%.

PORTFOLIO RISK : 2 – SECURITY CASE

$$\sigma_p = [w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1w_2 \rho_{12} \sigma_1 \sigma_2]^{\frac{1}{2}}$$

Example : $w_1 = 0.6$, $w_2 = 0.4$,

$$\sigma_1 = 10\%, \sigma_2 = 16\%$$

$$\rho_{12} = 0.5$$

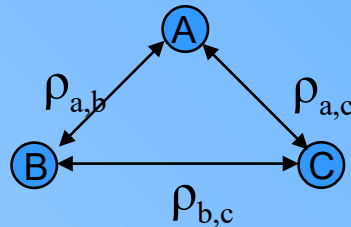
$$\begin{aligned}\sigma_p &= [0.6^2 \times 10^2 + 0.4^2 \times 16^2 + 2 \times 0.6 \times 0.4 \times 0.5 \times 10 \times 16]^{\frac{1}{2}} \\ &= 10.7\%\end{aligned}$$

■ The average standard deviation of two securities is 13, which is less than standard deviation of the portfolio, which is 10. Thus diversification reduces risk.

Risk of a Three-asset Portfolio

The data requirements for a three-asset portfolio grows dramatically if we are using Markowitz Portfolio selection formulae.

We need 3 (three) correlation coefficients between A and B; A and C; and B and C.



$$\sigma_p = \sqrt{\sigma_A^2 w_A^2 + \sigma_B^2 w_B^2 + \sigma_C^2 w_C^2 + 2w_A w_B \rho_{A,B} \sigma_A \sigma_B + 2w_B w_C \rho_{B,C} \sigma_B \sigma_C + 2w_A w_C \rho_{A,C} \sigma_A \sigma_C}$$

Three-Asset Portfolio Risk Matrix

	Security X_1	Security X_2	Security X_3
Security X_1	$X_1^2 \sigma_1^2$	$X_1 X_2 \rho_{12} \sigma_1 \sigma_2$	$X_1 X_3 \rho_{13} \sigma_1 \sigma_3$
Security X_2	$X_2 X_1 \rho_{21} \sigma_2 \sigma_1$	$X_2^2 \sigma_2^2$	$X_2 X_3 \rho_{23} \sigma_2 \sigma_3$
Security X_3	$X_3 X_1 \rho_{31} \sigma_3 \sigma_1$	$X_3 X_2 \rho_{32} \sigma_3 \sigma_2$	$X_3^2 \sigma_3^2$

P12 = correlation between security 1 and 2

P13 = correlation between security 1 and 3

P23 = correlation between security 2 and 3

PORTFOLIO RISK : 3 – SECURITY CASE

$$\sigma_p = [\sum \sum w_i w_j \rho_{ij} \sigma_i \sigma_j]^{1/2}$$

Example : $w_1 = 0.5$, $w_2 = 0.3$, and $w_3 = 0.2$

$$\sigma_1 = 10\%, \sigma_2 = 15\%, \sigma_3 = 20\%$$

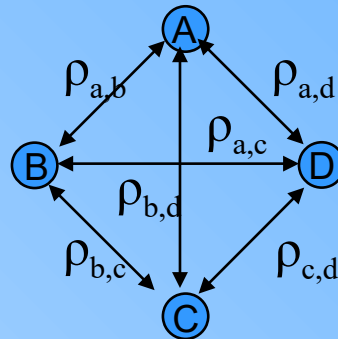
$$\rho_{12} = 0.3, \rho_{13} = 0.5, \rho_{23} = 0.6$$

$$\begin{aligned} \sigma_p &= [w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + w_3^2 \sigma_3^2 + 2 w_1 w_2 \rho_{12} \sigma_1 \sigma_2 \\ &\quad + 2w_2 w_3 \rho_{13} \sigma_1 \sigma_3 + 2w_2 w_3 \rho_{23} \sigma_2 \sigma_3]^{1/2} \\ &= [0.5^2 \times 10^2 + 0.3^2 \times 15^2 + 0.2^2 \times 20^2 \\ &\quad + 2 \times 0.5 \times 0.3 \times 0.3 \times 10 \times 15 \\ &\quad + 2 \times 0.5 \times 0.2 \times 0.5 \times 10 \times 20 \\ &\quad + 2 \times 0.3 \times 0.2 \times 0.6 \times 15 \times 20]^{1/2} \\ &= 10.79\% \end{aligned}$$

Risk of a Four-asset Portfolio

✓ The data requirements for a four-asset portfolio grows dramatically if we are using Markowitz Portfolio selection formulae.

✓ We need 6 correlation coefficients between A and B; A and C; A and D; B and C; C and D; and B and D.



Example

- Suppose Asset A has an expected return of 10 percent and a standard deviation of 20 percent. Asset B has an expected return of 16 percent and a standard deviation of 40 percent. If the correlation between A and B is 0.6, what are the expected return and standard deviation for a portfolio comprised of 30 percent Asset A and 70 percent Asset B?

Portfolio Expected Return

$$\begin{aligned}\hat{r}_P &= w_A \hat{r}_A + (1 - w_A) \hat{r}_B \\ &= 0.3(0.1) + 0.7(0.16) \\ &= 0.142 = 14.2\%.\end{aligned}$$

Portfolio Standard Deviation

$$\begin{aligned}\sigma_p &= \sqrt{W_A^2 \sigma_A^2 + (1 - W_A)^2 \sigma_B^2 + 2W_A(1 - W_A)\rho_{AB}\sigma_A\sigma_B} \\ &= \sqrt{0.3^2(0.2^2) + 0.7^2(0.4^2) + 2(0.3)(0.7)(0.4)(0.2)(0.4)} \\ &= 0.309\end{aligned}$$

RISK OF AN N - ASSET PORTFOLIO

$$\sigma_p^2 = \sum \sum w_i w_j \rho_{ij} \sigma_i \sigma_j$$

n x *n* MATRIX

	1	2	3	...	<i>n</i>
1	$w_1^2 \sigma_1^2$	$w_1 w_2 \rho_{12} \sigma_1 \sigma_2$	$w_1 w_3 \rho_{13} \sigma_1 \sigma_3$	...	$w_1 w_n \rho_{1n} \sigma_1 \sigma_n$
2	$w_2 w_1 \rho_{21} \sigma_2 \sigma_1$	$w_2^2 \sigma_2^2$	$w_2 w_3 \rho_{23} \sigma_2 \sigma_3$	...	$w_2 w_n \rho_{2n} \sigma_2 \sigma_n$
3	$w_3 w_1 \rho_{31} \sigma_3 \sigma_1$	$w_3 w_2 \rho_{32} \sigma_3 \sigma_2$	$w_3^2 \sigma_3^2$	...	
:	:				:
<i>n</i>	$w_n w_1 \rho_{n1} \sigma_n \sigma_1$				$w_n^2 \sigma_n^2$

Example 2

Suppose you have predicted the following returns on stock of X and Y company in four possible states of the economy as follows:

State of the economy	Probability	Return for stock X (Rx) (%)	Return for stock Y (Ry) (%)
Boom	0.25	18.5	15
Normal	0.25	10	11
Recession	0.25	2	6
Depression	0.25	-6	-4

Required:

Using the above data, calculate expected portfolio return and Risk if you invest 40% on stock X and 60% on Stock Y?

1. Find expected return of each security to find portfolio return

State	Pr.	(Rx) (%)	Ry (%)	PrRx	PrRy
Boom	0.25	18.5	15	4.625	3.75
Normal	0.25	10	11	2.5	2.75
Recession	0.25	2	6	0.5	1.5
Depression	0.25	-6	-4	-1.5	-1
sum				E(Rx)=6.13	E(Ry)=7

$$\begin{aligned} \text{➤ } E(R_x) &= 0.25(18.5) + 0.25(10) + 0.25(2) + 0.25(-6) \\ &= 6.13\% \end{aligned}$$

$$\text{➤ } E(R_y) = 0.25(15) + 0.25(11) + 0.25(6) + 0.25(-4) = 7\%$$

Cont....

2. Find standard deviation of each security

Pr.	R _x	R _y	(R _x -E(R _x))	(R _x -E(R _x)) ²	Pr (R _x -E(R _x)) ²	(R _y -E(R _y))	(R _y -E(R _y)) ²	Pr.(R _y -E(R _y)) ²
0.25	18.5	15	12.4	153.14	38.29	8	64	16
0.25	10	11	3.9	15.02	3.75	4	16	4
0.25	2	6	-4.13	17.02	4.254	-1	1	0.25
0.25	-6	-4	-12.125	147.02	36.754	-11	121	30.25
					var=83.045			Var=50.5

$$E(R_x)=6.13$$

$$E(R_y)=7$$

$$\sigma_x^2 = \sum p[R_x - E(R_x)]^2$$

$$\begin{aligned}\text{Var of } x = \sigma_x^2 &= 0.25(18.5-6.13)^2 + \\ & 0.25(10-6.13)^2 + \\ & 0.25(2-6.13)^2 + \\ & 0.25(-6-6.13)^2 \\ & = 38.2853 + 3.7539 + 4.25 + 36.75 = 83\end{aligned}$$

$$\text{St. D. of } x = \sigma_x = 9.11$$

$$\begin{aligned}\text{Var of } y = \sigma_y^2 &= 0.25(15-7)^2 + \\ & 0.25(11-7)^2 + \\ & 0.25(6-7)^2 + \\ & 0.25(-4-7)^2 \\ & = 16 + 4 + .25 + 30.25 = 50.5\end{aligned}$$

$$\sigma_y = 7.11$$

3. Find covariance between X&Y

State	Pr.	(Rx)	Ry	Deviation (Rx-E(Rx))	Deviation (Ry-E(Ry))	Product of deviation (Rx-E(Rx)) (Ry-E(Ry))	P(Rx-E(Rx)) (Ry-E(Ry))
Boom	0.25	18.5	15	12.375	8	99	24.75
Normal	0.25	10	11	3.875	4	15.5	3.875
Recession	0.25	2	6	-4.125	-1	4.125	1.03125
Depression	0.25	-6	-4	-12.125	-11	133.375	33.34375
							Cov=63

$$Cov_{xy} = \sum \{P_i [R_x - E(R_x)] [R_y - E(R_y)]\}$$

Cov xy= 0.25(18.5-6.13)(15-7.1)+
0.25(10-6.13)(11-7.1)+
0.25(2-6.13)(6-7.1)+
0.25(-6-6.13)(-4-7.1)=
=24.75+3.875+1.03125+33.34375=63

	Sec. X	Sec. Y
Exp. return	6.13	7
St. div.	9.11	7.11

RXY = $\frac{Cov_{xy}}{SD_x SD_y}$
63/(9.11*7.11) = 63/64.74=0.97

Cont..

	Security x	Security y
Exp. return	6.13%	7%
St. D	9.11	7.11
Weight	40%	60%
Correlation = -0.97		

Portfolio variance

$$\begin{aligned}\sigma_p^2 &= w_x^2 \sigma_x^2 + w_y^2 \sigma_y^2 + 2w_x w_y \text{COV}_{xy} \\ &= w_x^2 \sigma_x^2 + w_y^2 \sigma_y^2 + 2w_x w_y \sigma_x \sigma_y \text{Cor}_{xy}\end{aligned}$$

$$\sigma_p^2 = (0.40^2)(9.11^2) + (0.60^2)(7.11^2) + 2(0.40)(0.60)(63) = 61.7$$

$$\sigma_p = 7.8\%$$

Or

$$\sigma_p^2 = (0.40^2)(9.11^2) + (0.60^2)(7.11^2) + 2(0.40)(0.60)(9.11)(7.11)(-0.97) = 61.7$$

**THANK U
AND**